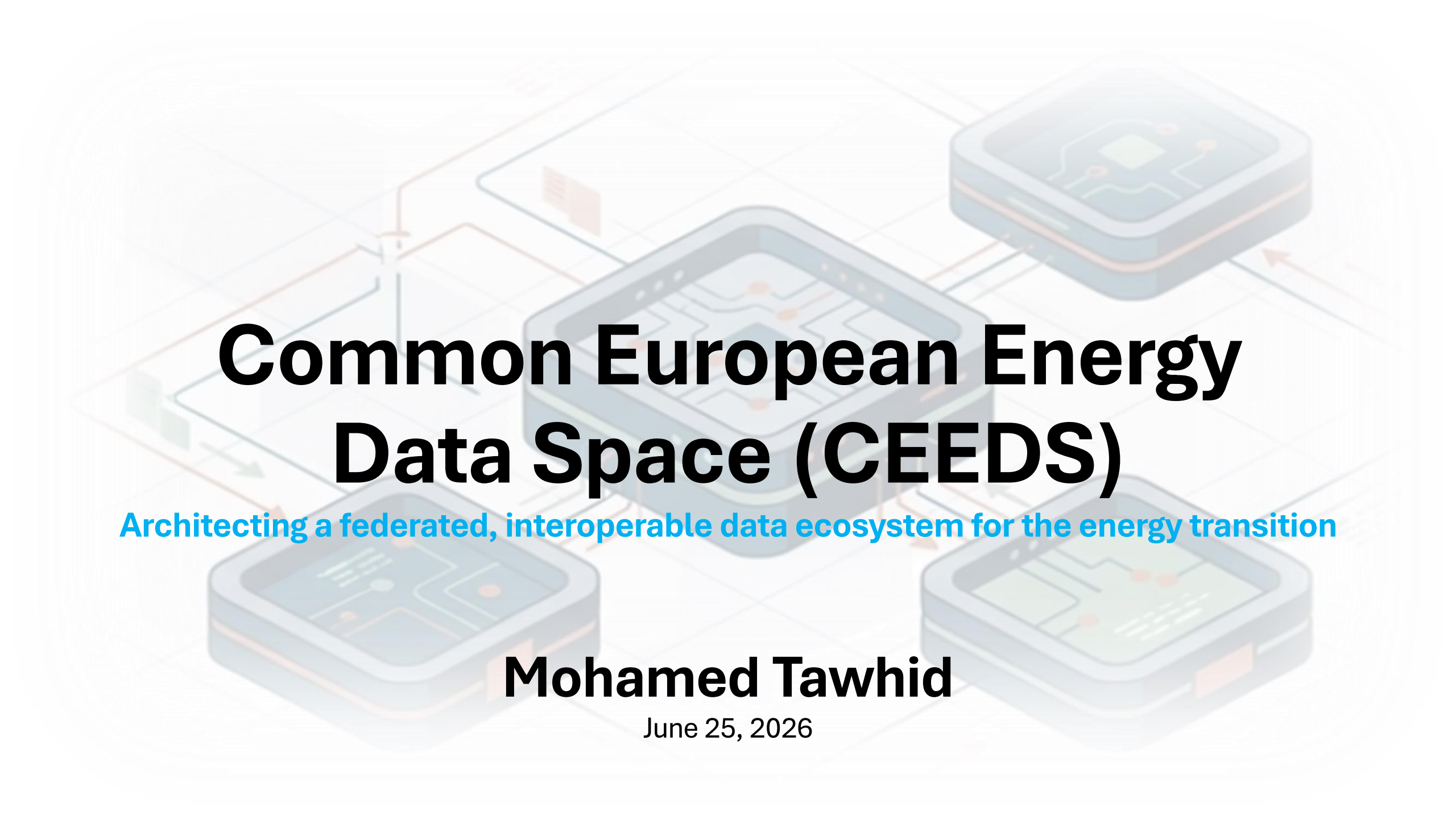




Common European Energy Data Space (CEEDS)

Architecting a federated, interoperable data ecosystem for the energy transition

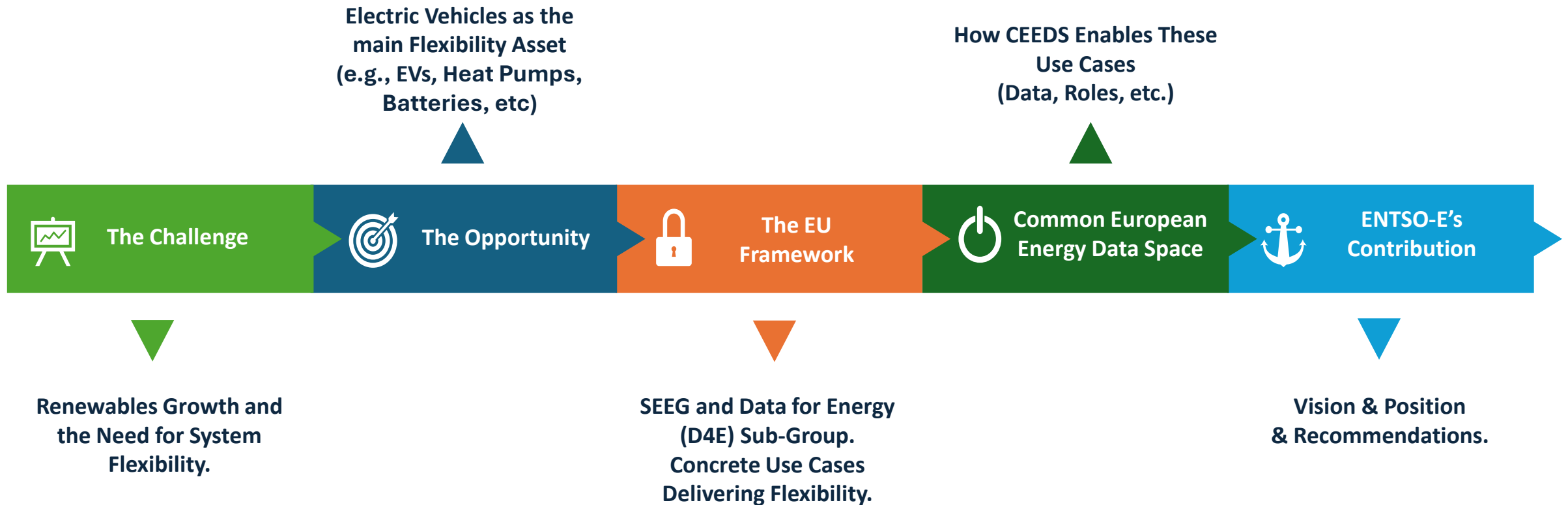
Mohamed Tawhid

June 25, 2026

Common European Energy Data Space (CEEDS)

Architecting a **federated, interoperable data ecosystem** to enhance the **Energy Transition**

Workflow





The Challenge that makes CEEDS necessary

Europe's Energy Challenge & Opportunity

International Energy Agency (IEA) & Eurostat

47% of the **European Union electricity** came from **renewables** in 2025

30% of the **EU electricity** came from **solar PV** and **wind**, surpassing the share of **fossil fuels** for the first time in 2025.

Key Message: **Variability is the new normal**

Leveraging **EVs' flexibility** to stabilize the grid requires **secure, real-time data exchange** across systems that speak different languages.

Distributed Flexibility (EV chargers & Batteries) exists but **TSOs cannot access it across borders** because:

1

Data is siloed by manufacturer, aggregator, and Member State.

2

No common semantic model for flexibility offers.

3

No trusted framework for cross-border data sharing.

Could Data Exchange Frameworks (CEEDS) be the solution for that challenge?



User Scenarios: Three EV Charging Realities (Context-specific Role Models)

The User Scenario Groups – Why Context Changes Everything

Roles and responsibilities shift dynamically based on:
Property Ownership, Grid Connection, and Charging Infrastructure Configuration.

100% Public (On Route/Highway)



- **Characteristics:** CPO owns both charger and grid meter.
- **Focus:** Fast charging (no V2G).
- **Key Actors:** CPO, MSP.

Semi-Private/Public (At Work/Multi-Family)



- **Characteristics:** Charger and grid meter have different owners.
- **Focus:** Multi-party smart charging; Delegated authority.
- **Key Actors:** EVSO, CEM, DER-A.

100% Private (Individual Home)



- **Characteristics:** Private citizen controls wallbox and metering point.
- **Focus:** Full bidirectional V2H/V2G potential.
- **Key Actors:** Final Electricity Customer, FSP.

Abbreviation Sheet: [Abbreviation Sheet_Joint Recommendation report.xlsx](#)

Semi-private/Public Charging User Scenario

Navigating EV Charging Scenarios: The “At Work” Framework

Defining primary charging categories and detailing smart charging roles in workplace environments.

Primary EV Charging Scenarios



All Public Charging

A Charging Point Operator (CPO) owns both the charging point and the grid-connected metering point.

Semi-private / Public Charging

Features delegated authority where a property owner utilizes a 3rd party for charging services.



All Private Charging

A private individual owns and operates both the charging point and the metering point.



Inside the “At Work” Scenario

Focusing on multi-party coordination for energy management and grid flexibility.

Secure Authentication

MSPs authenticate vehicles on behalf of Fleet Owners to enable secure charging access.



Smart Optimization Roles

DER Aggregators and FSPs optimize charging to provide market access and building energy management.



Balanced Charging Schedules

Coordination is required when individual EV driver needs conflict with property owner energy boundaries.



Sub-categories within Semi-private/Public

Scenario Category	Typical Locations	Key Characteristic
 At Work	Workplaces, Depots	Balancing fleet and employee needs
 On Route	Public destination points	Charging while traveling to non-home destinations
 At Home	EV End User residence	Charging via delegated property authority

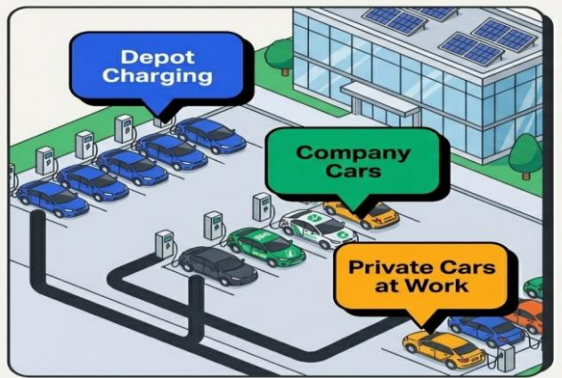


Walk through the At-Work Scenario in detail

Roles Mapping: The Alignment Framework maps Functional Roles across Domains

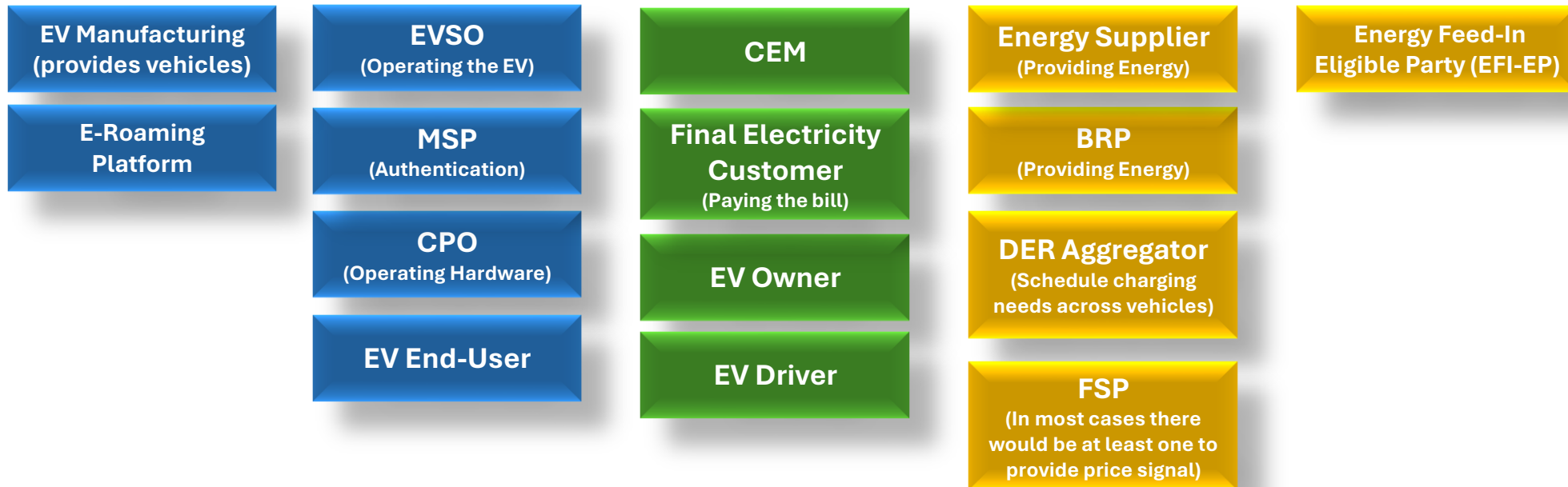
User Scenario at Work

'At Work' scenario introduces multi-party complexity

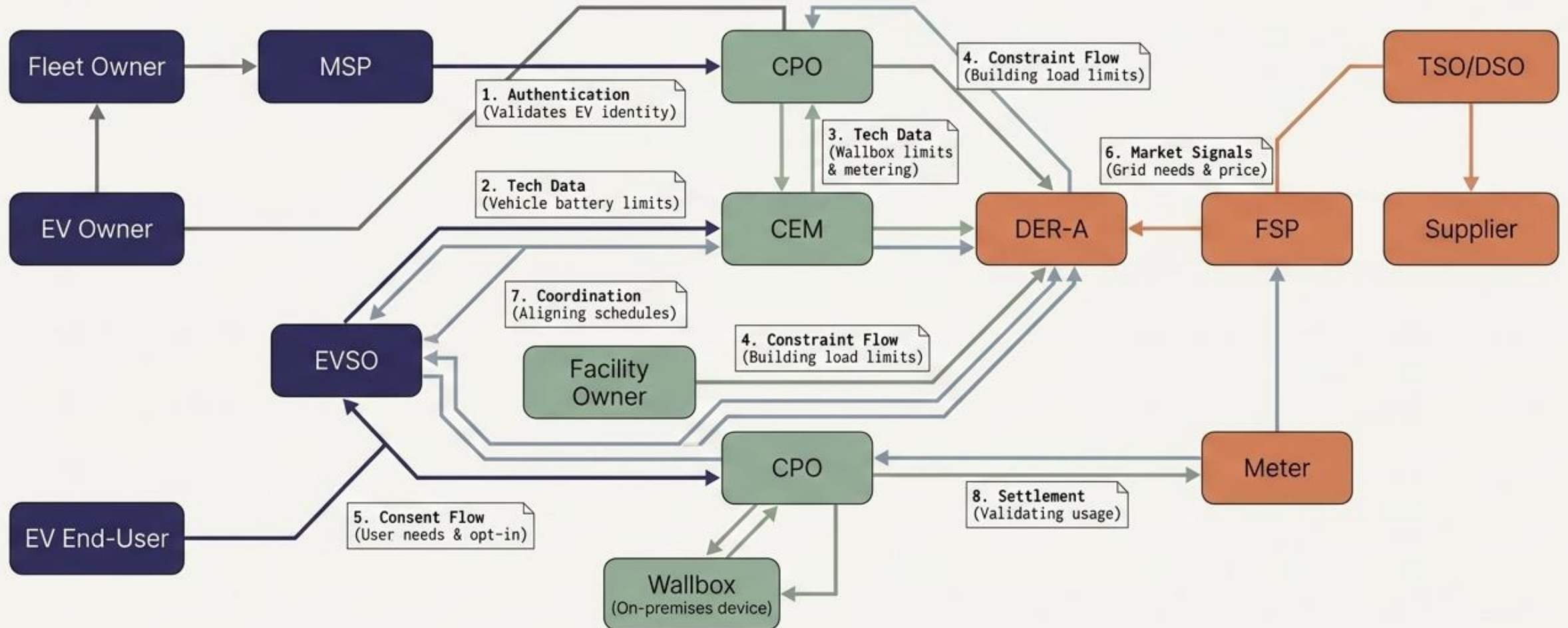


The **semi-private / at work scenario** is the **most complex** and **most representative** of real-world flexibility **challenges**. Thus, it serves as the **stress test** for the entire **data exchange framework**.

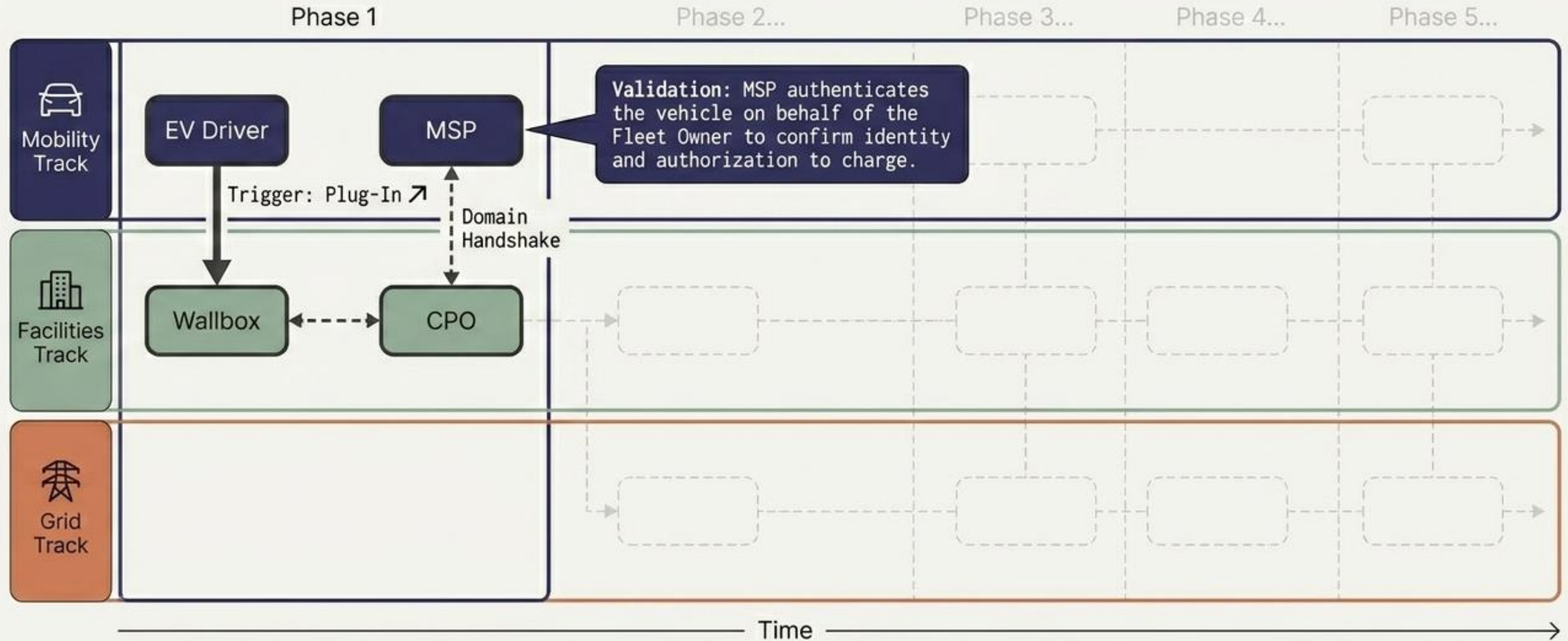
Multiple sectors must collaborate to enable smart charging



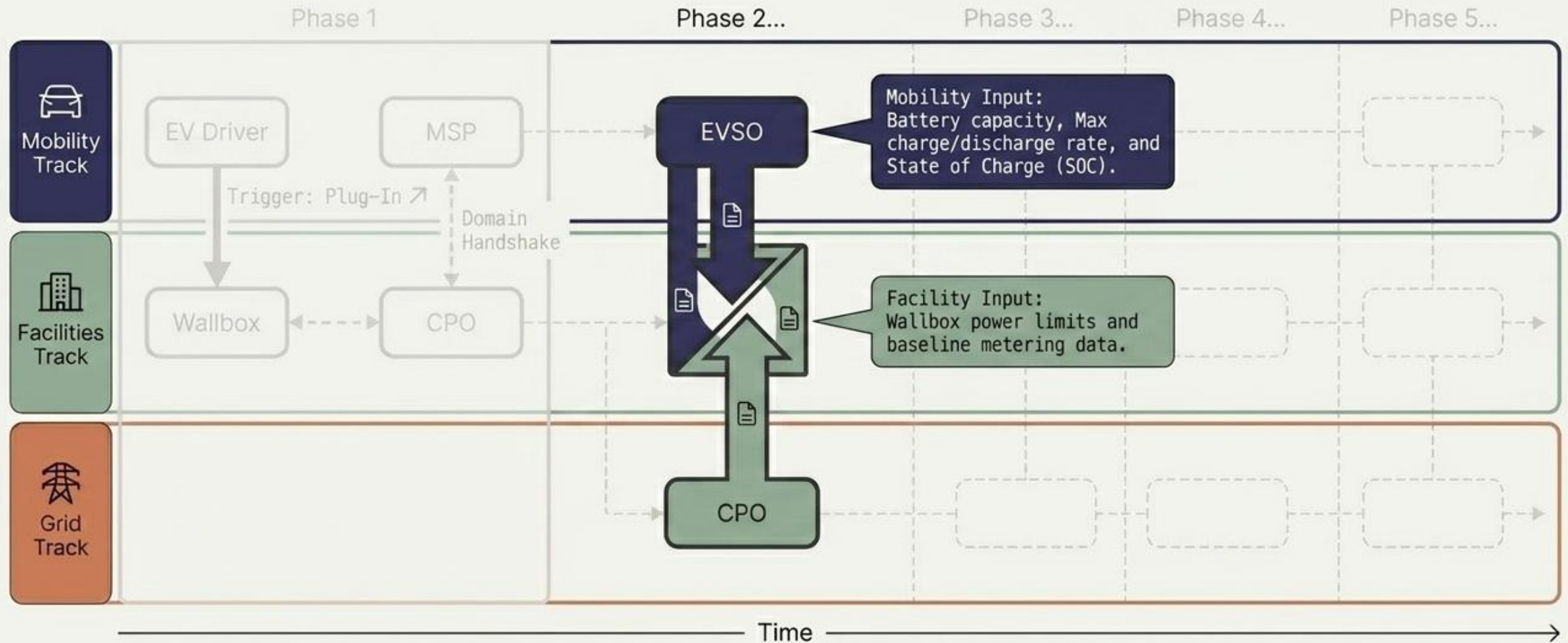
Mapping the critical interfaces and data flows across the ecosystem



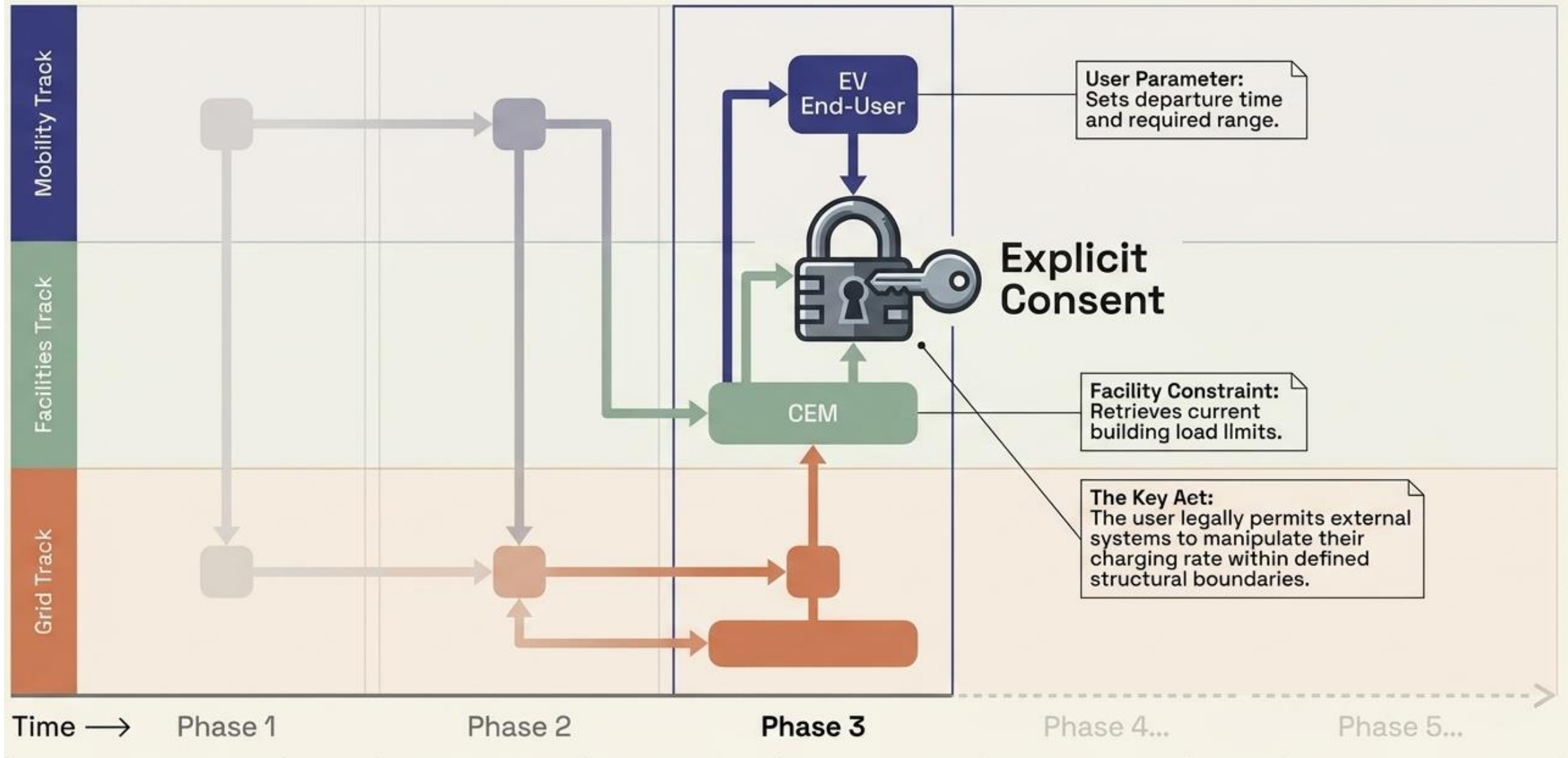
Phase 1: Establishing identity and authorization at the plug



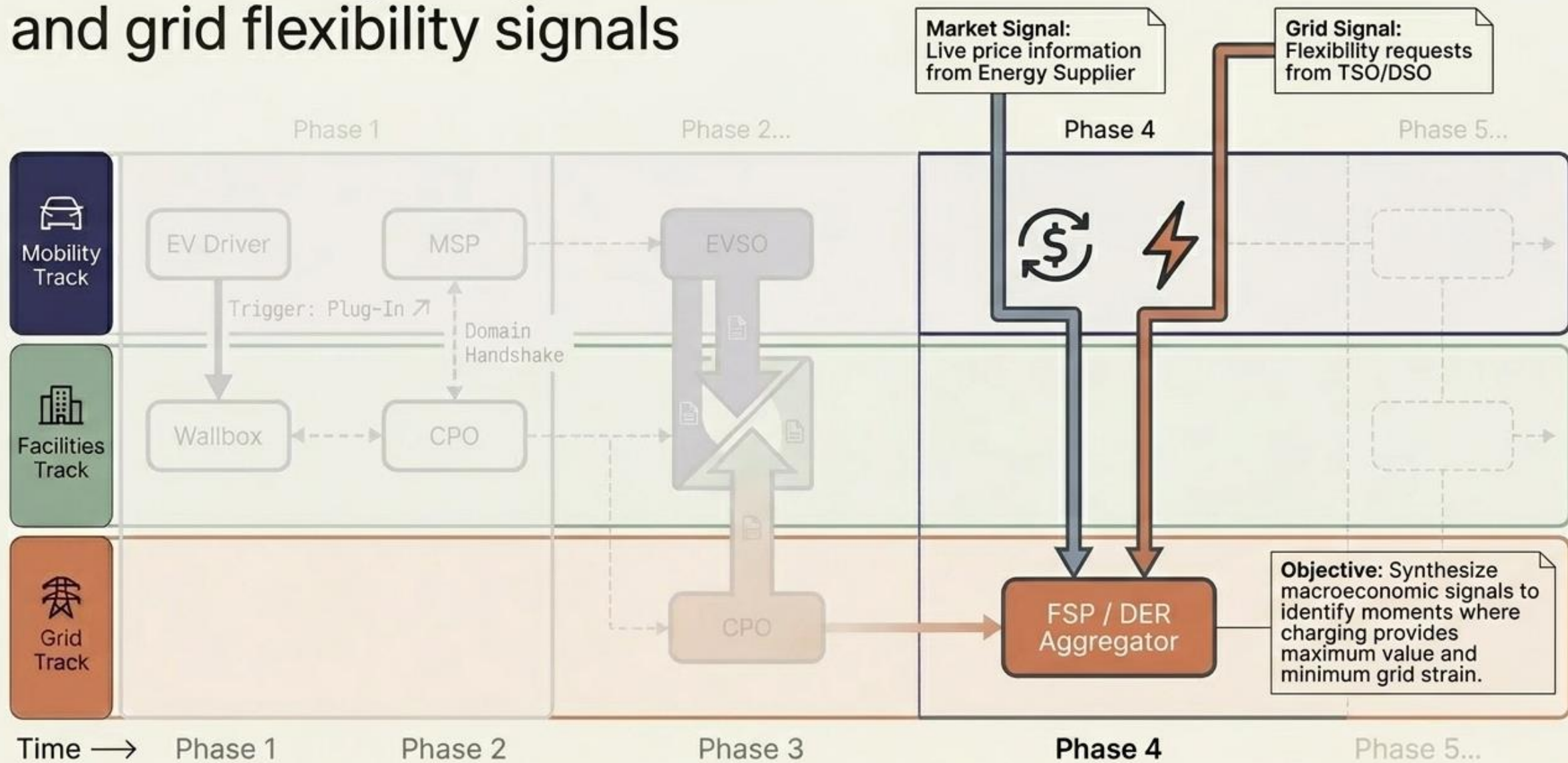
Phase 2: Defining the technical boundaries of the vehicle and infrastructure



Phase 3: Aligning facility constraints with end-user consent

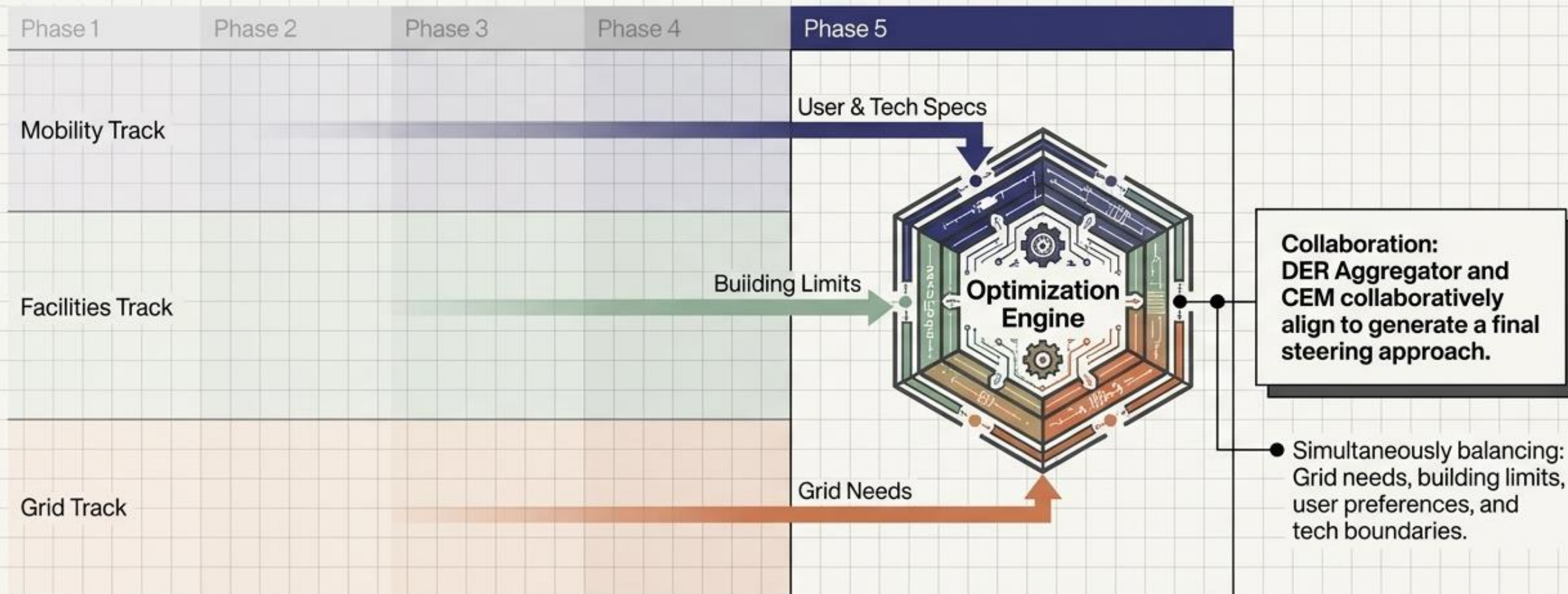


Phase 4: Integrating external market and grid flexibility signals



Phase 5: Generating the optimized smart charging schedule

Time →

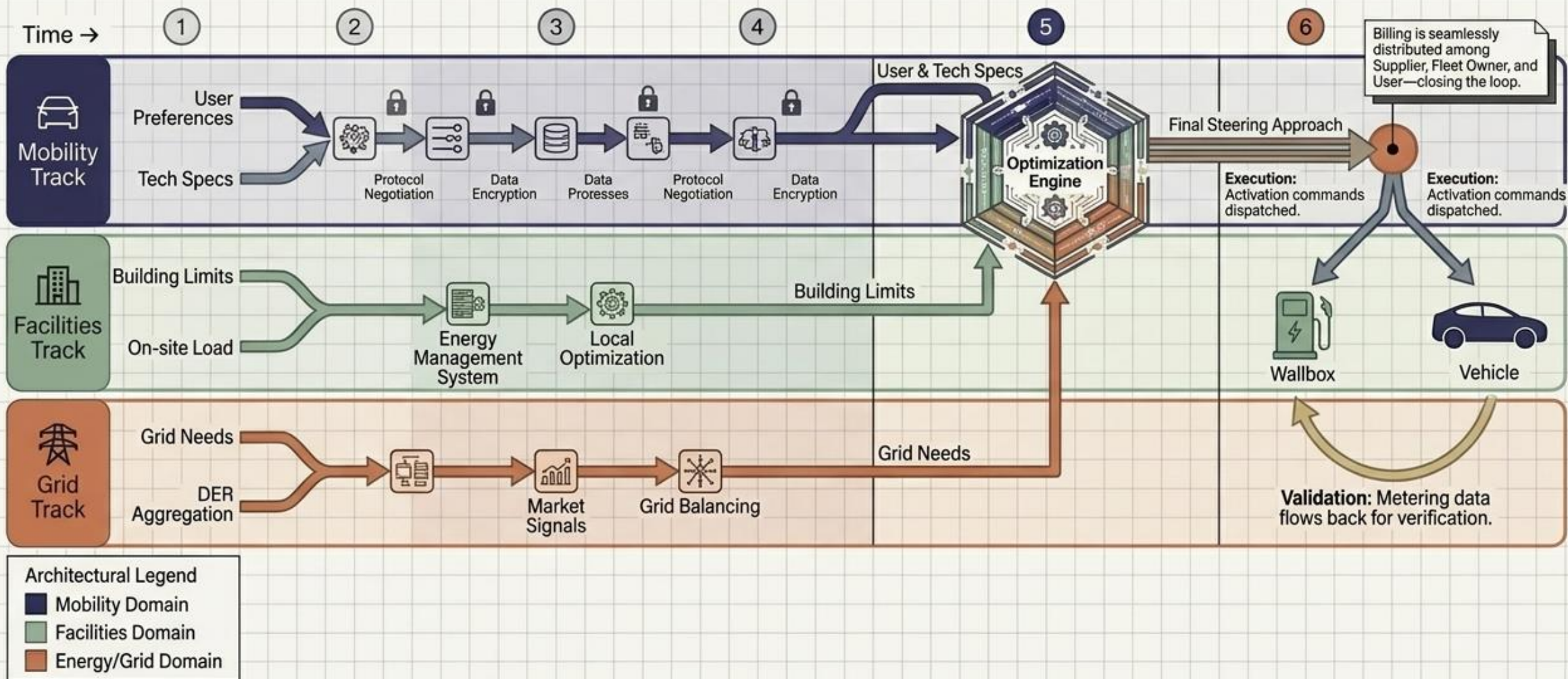


Phase 6: Executing dispatch and validating settlement

Time →



The complete multiparty orchestration lifecycle



From the scenarios, the D4E-STF-CoW joint work group derived 4 Priority Use Cases that define the end-to-end data flow for flexibility

The Four Priority Use Cases

D4E-STF-CoW – Joint work group on data exchange for flexibility and smart and bidirectional charging

Use Case	Name	Scope	Connects to
UC1	Flexibility information system (registration, prequalification, verification) and flexible connection agreements.	Flexibility & Congestion management – grid operator identifies need and procures flexibility.	Triggers UC2; uses UC4 for validation
UC2	Flexibility market operation (e.g. offers, bids) and flexibility settlement .	Pan-European & Local Platforms – market platforms for balancing and local flexibility trading .	Receives from UC1; sends to UC3
UC3	Activation (explicit and implicit) of distributed energy resources .	FSP/DER Aggregator/DER Operator – bidding, portfolio optimisation, activation .	Bids into UC2; activates DERs via UC4
UC4	Data exchange from smart meters , dedicated measurement devices and energy management systems.	On-Premises Device & Asset-level data – real-time site/device data, consent, activation .	Feeds data to UC3; receives commands from UC1/UC2

Core Flow: Grid need (UC1) → Market platform (UC2) → FSP/Aggregator bid (UC3) → On-prem activation (UC4) → Measurement back up

STF: Sustainable Transport Forum -- **CoW:** Coalition of the Willing on Smart and Bi-Directional Charging – **FSP:** flexibility Service Provider -- **DER:** Distributed Energy Resource



CEEDS Vision that enables these Use Cases

Common European Energy Data Space (CEEDS) (The Vision & Framework)

CEEDS: Federating Europe's Energy Data for Interoperability

1

The “WHAT”: Core Definition

CEEDS is an **EU initiative** creating a common framework to **federate national** and **regional** energy platforms across Europe. It is **not a central database**, Instead, it is a **decentralised federation** of independent **data spaces** where each participant **retains control** over its data. This enables secure, sovereign data sharing across **electricity** and **e-mobility sectors**.

Ex: The European Health Data Space & The European language data space.

2

The “WHY”: Main Objective

The **goal** is to **break down data silos** to enhance efficiency, reliability, and sustainability in the energy sector. It enables new **data-driven services** to **balance the grid**, **integrate renewables**, and **accelerate decarbonization**.

3

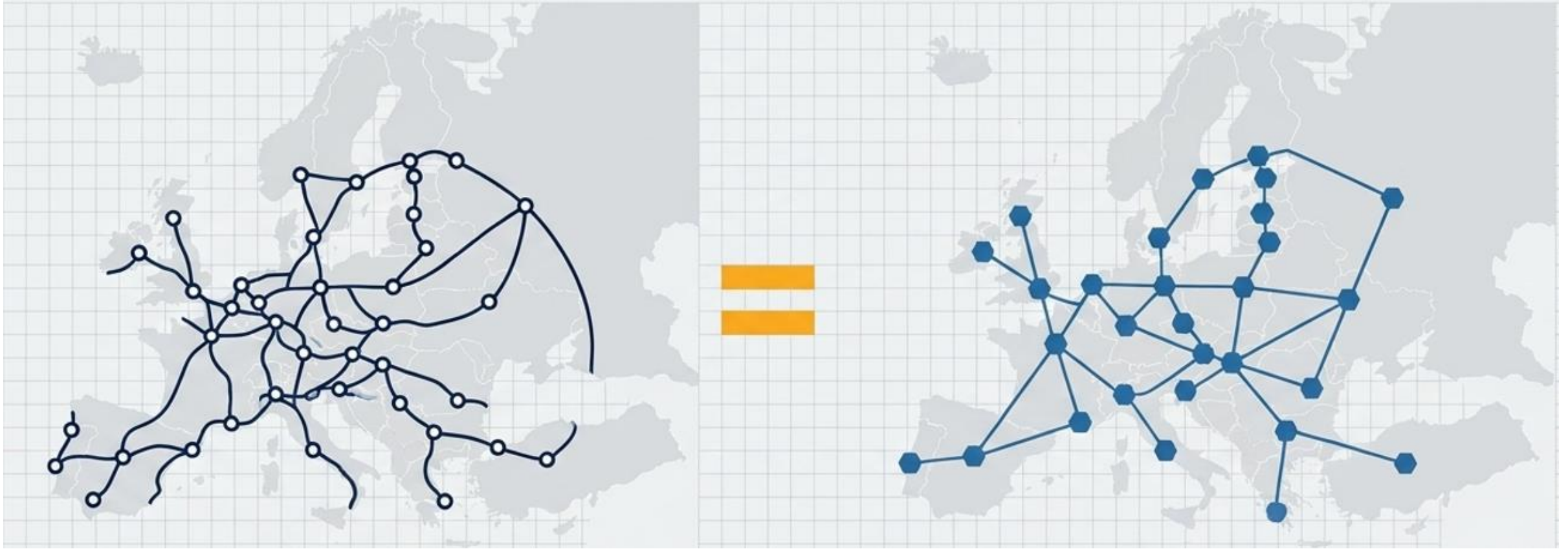
The “HOW”: Governance Model

Operationally, CEEDS is about establishing a governed community. It **defines** clear **rules, policies**, and **membership criteria** for participants, all managed by a designated Data space governance framework that **ensures compliance** and **trust** across the federated ecosystem.

What is a Data Space?



Infrastructure Analogy

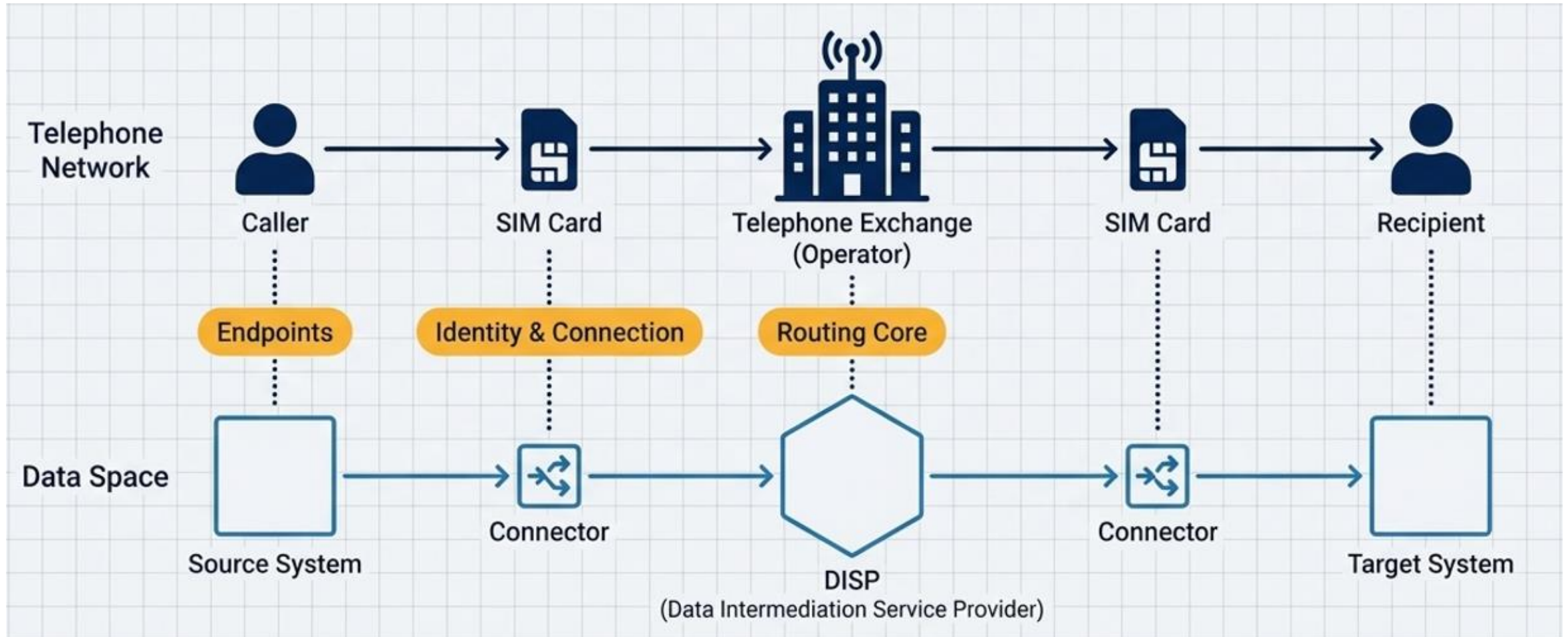


The Pan-European Telephone Network enabled seamless continent-wide human connectivity.

Common European Data Spaces enable seamless continent-wide information system connectivity.

Intermediation Services Route Data Identically to Telephone Exchanges

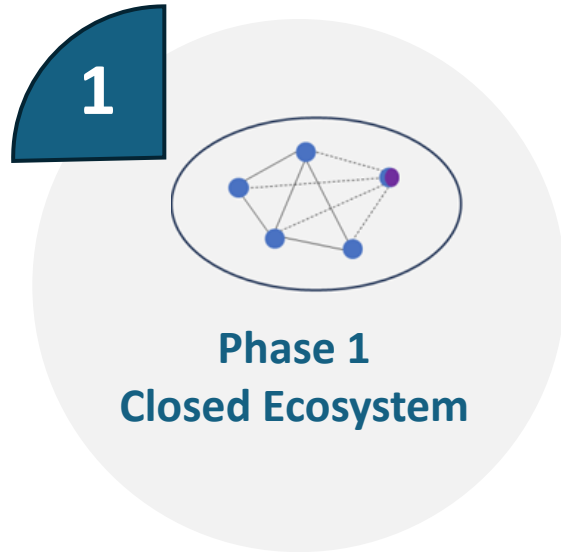
As the **telecom network** scales through interconnected operators, the **Data Space** is created by networking authorized DISPs.



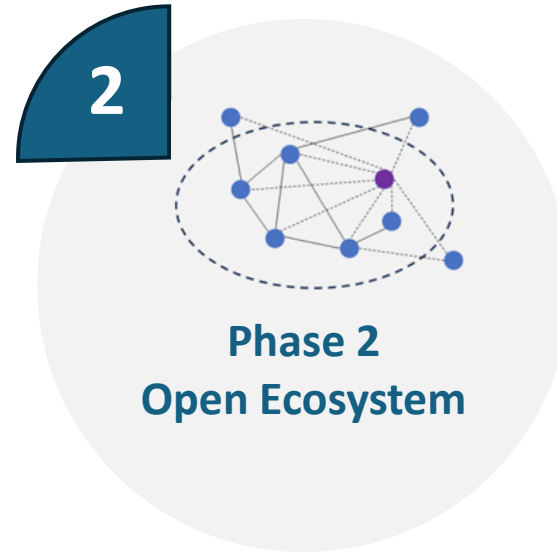
How to securely exchange data in a European Data Space



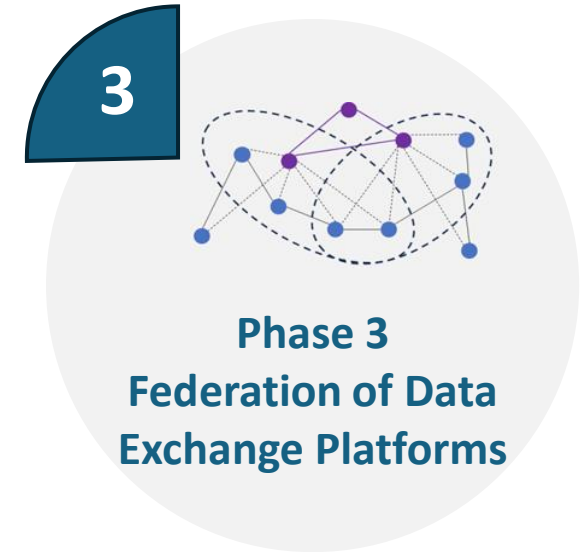
The Evolution Toward Federated Data Exchange



Isolated energy platforms with restricted, bilateral data sharing.



Standardized APIs allow broader data exchange but remain limited by centralized governance.



Interconnected, sovereign platforms exchanging data and metadata across European borders.

Legend:

- **Roles:** ● Participants (data provider / data consumer); ● Federator
- **Data exchanges:** — Payload data (incl. metadata between participants)
- - - Metadata between participants and federator
— Metadata between federators
- **Ecosystems/Platforms:** ○ Closed; ○ Open

What kind of Data Shared?

Market Data: Flexibility requests, unique bid IDs, price information, and activation signals.

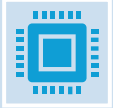
Technical & Metering Data: Real-time telemetry, validated historical metering data, and **baselines** (the reference profile used to calculate how much energy was actually shifted).

Edge Device Data: Information on specific appliance energy use from **Demand Management Devices (DMDs)** and overall site consumption from **smart meters**.

Governance Data: Customer consent records, registration statuses, and prequalification results.

Static & Structural Data: Asset characteristics (e.g., **battery capacity**, location), party roles and identifiers, contract terms, eligibility criteria.

The Values CEEDS Brings



Interoperability & Standardisation: Replaces fragmented national standards with a unified European **framework**, lowering integration costs for manufacturers and service providers.



Grid Stability: Enhances resilience by allowing **grid operators** to **coordinate** decentralized assets to **prevent congestion** and **integrate** more **renewables**.



Consumer Empowerment: Prosumers gain "**data sovereignty**," using consent-based **tools** to **participate** in **new markets** and earn revenue from their flexible assets.



Market Transparency: Ensures that **flexibility** is **quantified** and settled accurately, **building trust** in the **reliability** of decentralised energy services.

ENTSO-E Strategic Objectives

1. **DECARBONISATION:** CEEDS unlocks EV flexibility for renewable integration.
2. **SECURITY:** CEEDS ensures secure, interoperable, TSO-ready data exchange.
3. **AFFORDABILITY & COMPETITIVENESS:** CEEDS enables a single market for flexibility, lowering costs.





ENTSO-E's Contributions & Recommendations

ENTSO-E Contributions to the Joint Report – Addressing Risks with Recommendations

	Joint Report Content / Issue	Risk	ENTSO-E Recommendation to Mitigate Risk
1	Standardisation – CEEDS should adopt existing standards (CIM/ESMP) rather than reinventing them.	Without a clear mandate, work may duplicate or conflict with JWG-TF3, leading to incompatible national standards that undermine TSO stability.	Issue a clear mandate to CEN/CENELEC/IEC under EU Regulation to adopt CIM/ESMP for CEEDS.
2	Governance & Clarity – The governance model, responsibilities, and transition path are not fully defined.	Ambiguity may cause delays or fragmented implementation across European countries.	Request explicit guidance from the European Commission on governance and transition steps.
3	Roles – Undefined roles (e.g., Energy Feed In Eligibility Party (EFI-EP)) and confusion between DER Aggregator (market-facing) and DER Operator (technical control).	Continued ambiguity complicates cross-sector flexibility and misaligns mobility-energy role definitions.	Remove undefined roles ; clearly separate DER Aggregator from DER Operator in the role model.
4	CEEDS Scope – A “data-first” rather than “problem-driven” approach.	CEEDS may become too abstract; member countries may develop incompatible local data spaces , fragmenting the flexibility market.	Adopt a problem-driven scope – first identify concrete challenges, then define data flows.

Key message on Standardisation:

DIA WG experts currently work on an alignment between existing standardisation bodies and specifically to promote CIM/ESMP

Comparison between Previous Initiatives for Data Spaces

Component	EDDIE	Synergies	Data Cellar	Omega-X	Enershare
Connector	EDDIE Framework	Synergies Connector	Eclipse Dataspace Components (EDC)	Eclipse Dataspace Components (EDC)	TNO Security Gateway
Identity Management	Keycloak & eID	Keycloak OIDC	OID4VC & Gaia-X	IDSA DAPS & Gaia-X	Keycloak & TSG
Contracting	Data Marketplace	DLT Smart Contract	IDS Contract Protocol	Gaia-X Federator	Blockchain ERC20
Publication & Discovery	Master Data Model	Data Marketplace	Gaia-X Federated Catalogue	Gaia-X Federated Catalogue	IDS Metadata Broker

Conclusions



Next Steps

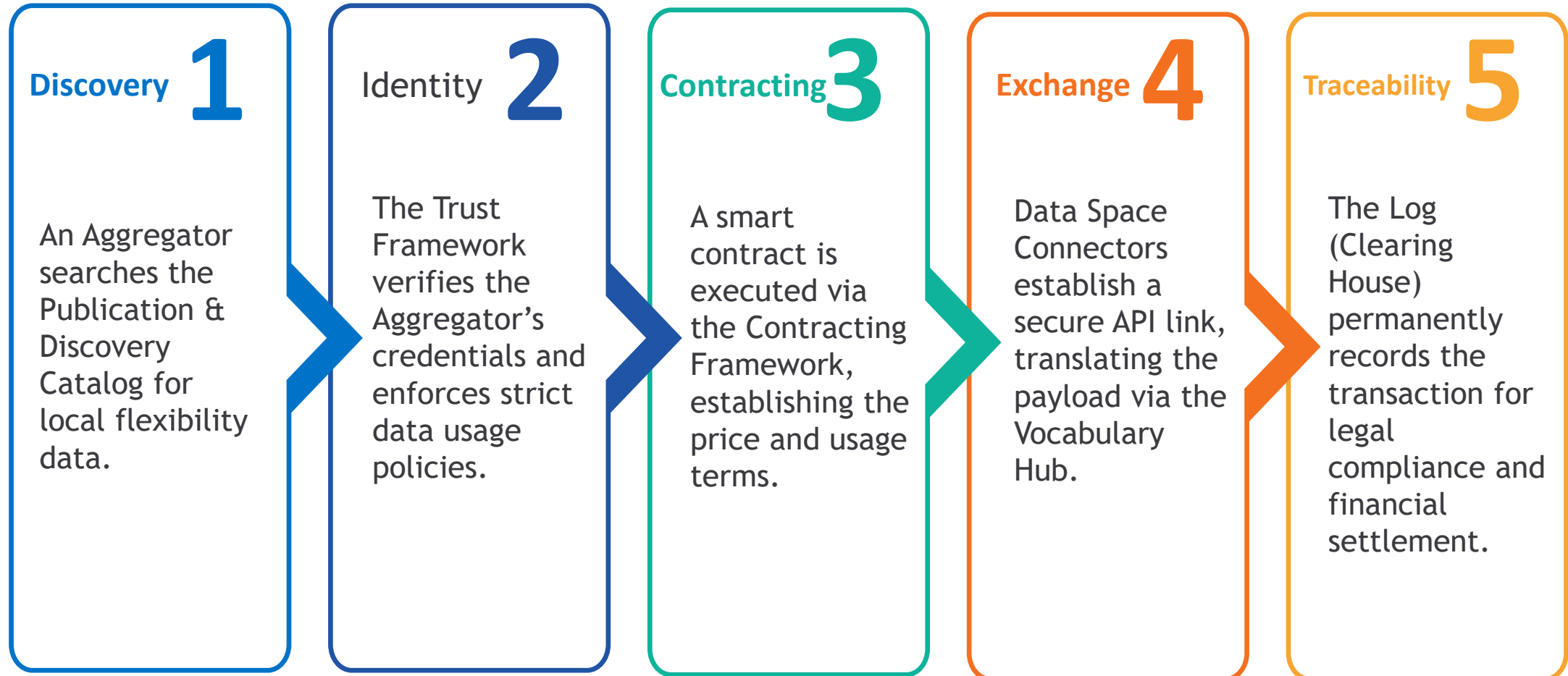
- Following the expert-level meeting in The Hague (2-3 June), the European Commission will continue advancing the development of the Common European Energy Data Space.
- Open topics are now being addressed in newly established subgroups, where DIA WG experts will remain actively involved. Current open topics: Regulatory Hurdles, Priority Use Cases, Common TSO-DSO signal, Testing and Verification.
- This is a long-term initiative with no fixed delivery timeline at this stage.
- Next steps include a new expert-level meeting planned for November.



Thank you!
Q&A

Synthesis: The Federated Architecture in Action

Seamless Pan-European Data Sovereignty Maintained Throughout



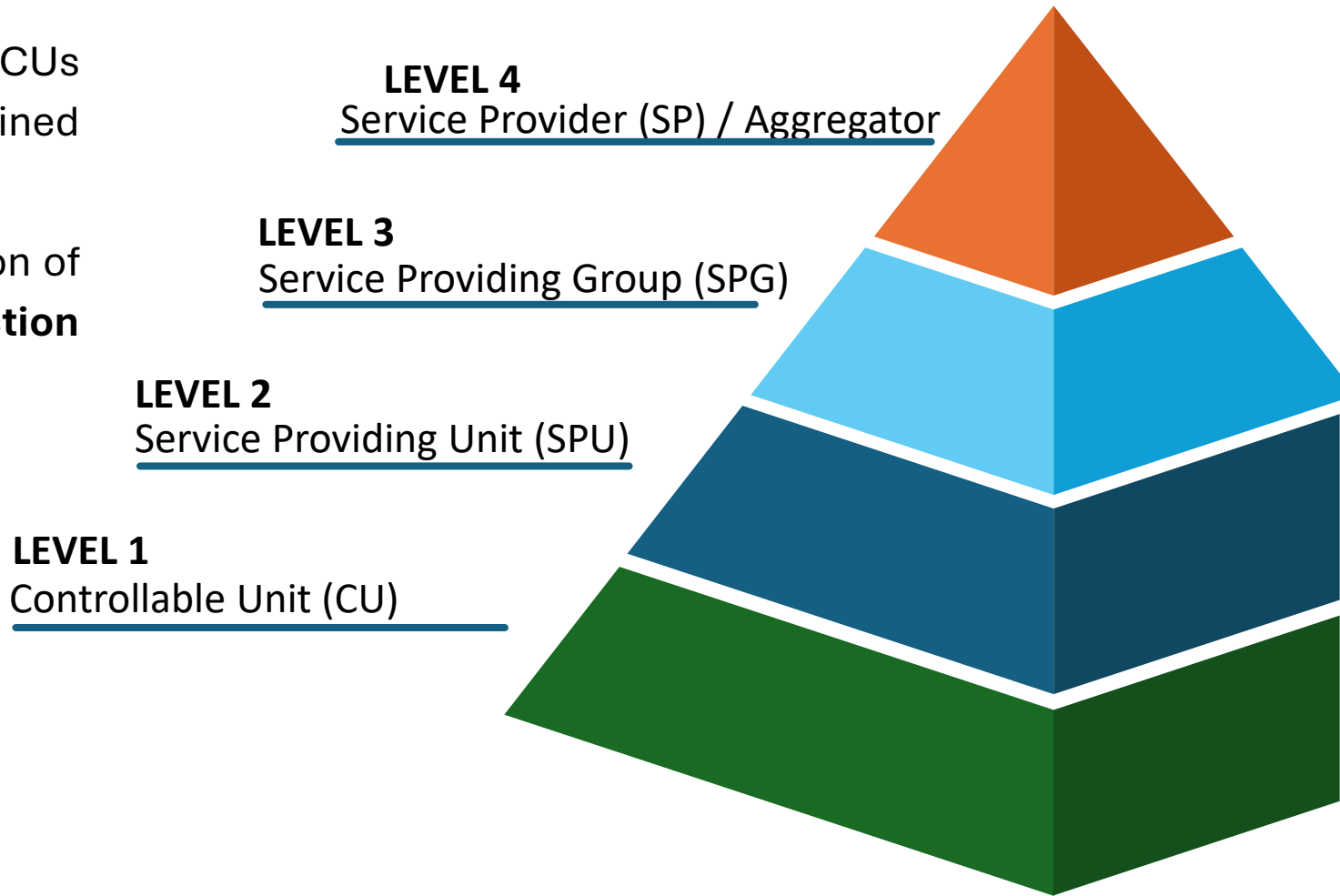
Summary of Role Interactions (Key Interfaces)

Interaction	Roles Involved	Purpose
Authentication	Fleet Owner ↔ MSP ↔ CPO	Validate the EV's identity and right to charge.
Technical Data Flow	EVSO ↔ CEM / DER-A	Provide vehicle battery limits.
Technical Data Flow	CPO ↔ CEM / DER-A	Provide wallbox power limits and metering.
Constraint Flow	Facility Owner / CEM ↔ DER-A	Provide building load limits and delegation rules.
Consent Flow	EV End-User ↔ CEM / DER-A	Provide user needs and opt-in for flexibility.
Market Signal Flow	TSO/DSO / Energy Supplier ↔ FSP / DER-A	Provide grid need and price signals.
Coordination & Steering	EVSO ↔ CPO ↔ CEM ↔ DER-A	Align on the final charging schedule and dispatch commands.
Settlement	CPO / Meter ↔ Energy Supplier / FSP	Validate usage and finalize billing.

Regulatory Taxonomy:

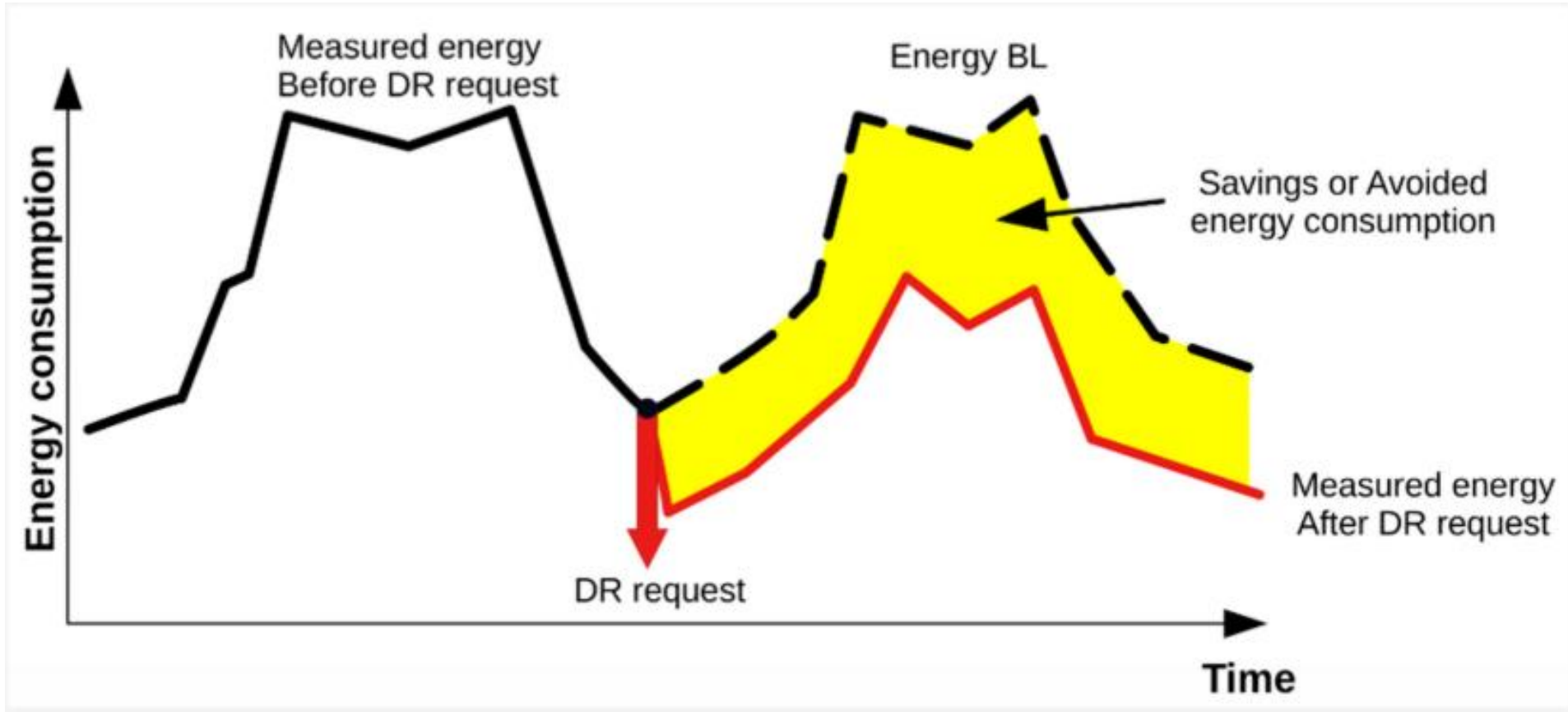
Nested Aggregation Structure

1. **Controllable Unit (CU):** A single power-generating module or demand unit. "Small CU" Maximum continuous active power $\leq 50\text{kW}$.
2. **Service Providing Unit (SPU):** An ensemble of CUs connected to a **single connection point**. Defined by the provider to offer services.
3. **Service Providing Group (SPG):** An aggregation of CUs or SPUs connected to **multiple connection points** within the same scheduling area.
4. **Service Provider (SP) / Aggregator:**
Manages the SPG to bid into markets.



Baseline & Metering

Counterfactual reference used to measure delivered service.

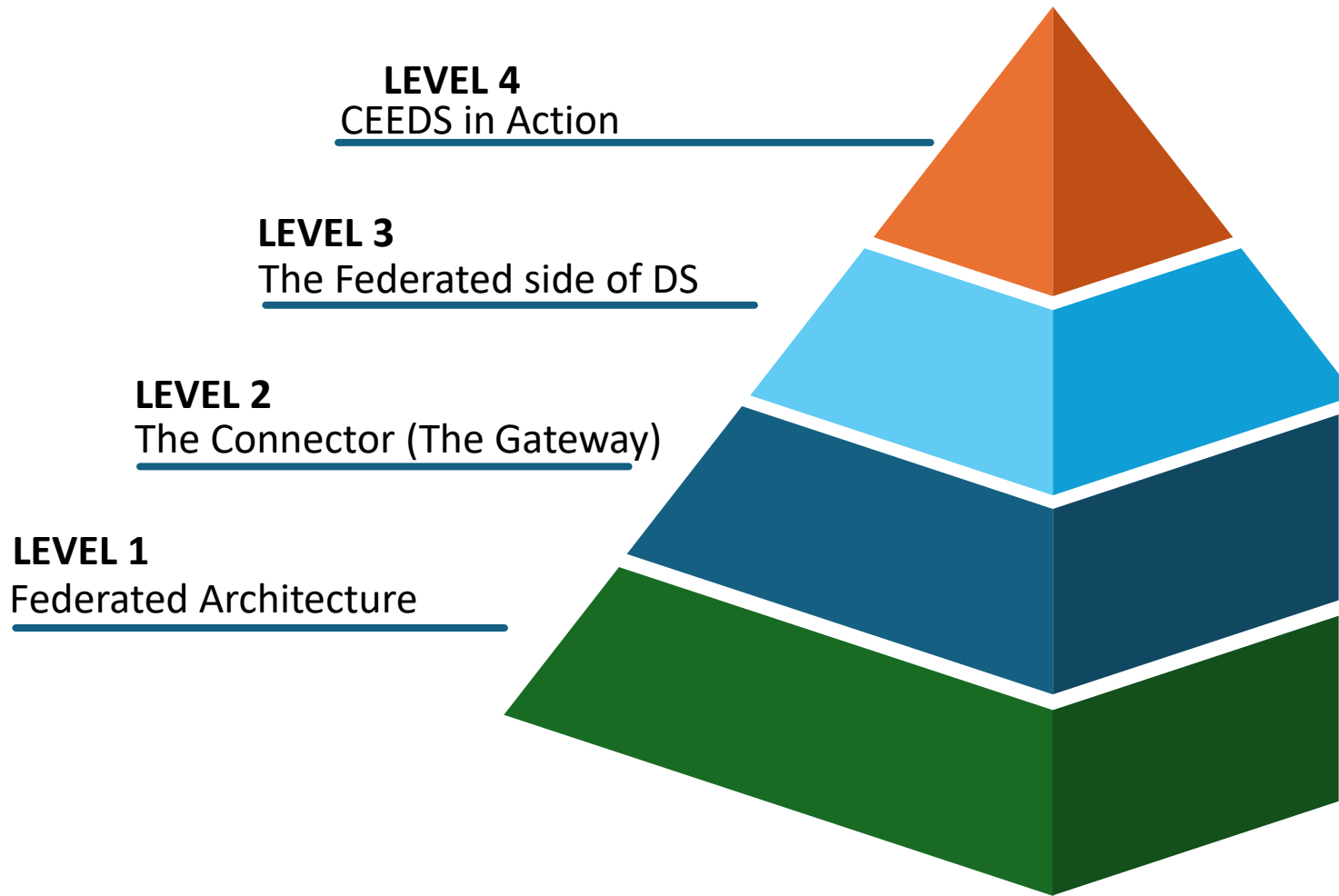


Example of a DR event with a measured energy balancing. The yellow area represents the avoided energy consumption after the DR request.

CEEDS Technical Architecture

A Federated, Decentralized Approach to Data Sharing

1. **Federated Architecture:** CEEDS uses a decentralized configuration, where data stays with its provider (e.g., a grid operator) rather than being copied to a central repository. This respects data sovereignty and reduces duplication.
2. **The Connector (The Gateway):** The fundamental building block is a piece of software called a data space connector, installed at each participant's site. It acts as a secure gateway for all communication, enforcing usage policies and managing actual data transfer.
3. **Key Components:** The federated "side" of the data space includes:
 1. **Trust Framework:** Manages identity, credentials, and access rights.
 2. **Vocabulary Hub:** Ensures shared semantic understanding of energy data.
 3. **Contractual Framework & Catalog:** Defines and stores machine-readable usage contracts and lists available data products.



How CEEDS Works Technically

Enabling Secure, Interoperable Data Exchange

Technical Interoperability: CEEDS aims for **interoperability at the technical, semantic, and governance levels**. This ensures different systems can "speak the same language" for seamless data exchange.

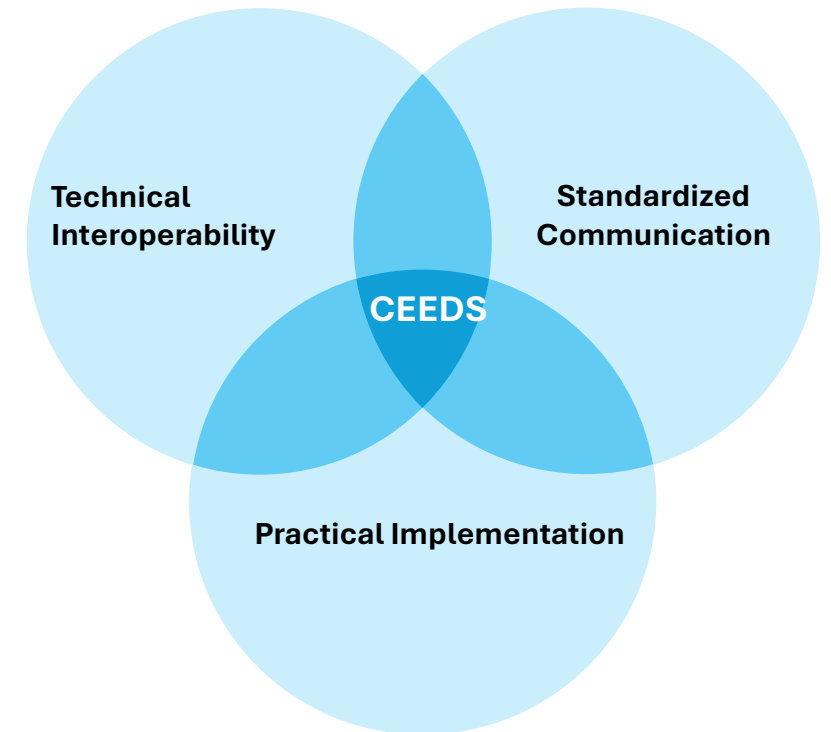
Standardized Communication: Connectors communicate using standardized protocols and APIs (like **ETSI NGSI-LD**) for core tasks such as:

- Discovering** available data assets.

- Negotiating** data usage contracts.

- Executing** secure, policy-enforced data transfers.

Practical Implementation: Concrete business use cases like **e-mobility roaming, load forecasting, and TSO-DSO coordination**, demonstrating how the architecture supports real-world energy sector challenges.



Comparison with Traditional Data Architectures

Feature	Data Space	Data Lake / Warehouse
Architecture	Decentralized/Federated: Data remains with the provider.	Centralized: Data is aggregated into a single repository.
Control	Sovereign: The data owner retains full control and defines usage policies.	Centralized Control: The platform owner manages access and governance.
Primary Focus	Secure, cross-organizational data sharing and collaboration.	Internal analytics , reporting, and data science on aggregated data.
Trust Model	Distributed Trust: Based on verified identities, contracts, and policy enforcement.	Single Point of Trust: Relies on the central platform's security and governance.
Data Movement	Data is transferred directly between participants only as needed (Peer-to-Peer).	Data is constantly moved or copied into the central system for storage and analysis.